

Speech Emotion Recognition Models

Capabilities, Outputs, and Documentation

Overview

This document outlines the capabilities, outputs, and foundational resources for three prominent AI models utilized in speech emotion recognition (SER) and audio event detection.

1 SenseVoice (FunAudioLLM)

SenseVoice is part of the FunAudioLLM family (developed by Alibaba), an audio foundation model that handles multilingual speech recognition, emotion recognition, and audio event detection natively without needing a separate classifier [1].

- **Official Paper:** FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs
- **GitHub Repository:** FunAudioLLM / SenseVoice
- **Model Weights:** SenseVoice-Small on Hugging Face

2 Whisper (OpenAI & Fine-Tuned SER)

Whisper is a highly robust, general-purpose speech recognition model trained on 680,000 hours of multilingual, weakly supervised data [3]. While OpenAI's base model only outputs text, the open-source community has heavily fine-tuned it for Speech Emotion Recognition (SER).

- **Official Paper:** Robust Speech Recognition via Large-Scale Weak Supervision
- **Official GitHub (Base Model):** OpenAI / Whisper
- **Fine-Tuned SER Repository (Hugging Face):** Whisper Large V3 Fine-Tuned for Emotion

3 Wav2Vec 2.0 (Meta & IEMOCAP Fine-Tuned)

Wav2Vec 2.0 revolutionized speech processing by introducing a self-supervised learning framework that masks speech input in the latent space, requiring significantly less labeled data for downstream tasks [2]. It is one of the most popular base models to fine-tune on the IEMOCAP dataset for detecting conversational frustration and sarcasm.

- **Official Paper:** wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations
- **Official GitHub (Base Model):** FacebookResearch / fairseq (Wav2Vec 2.0)
- **Fine-Tuned IEMOCAP Repository (Hugging Face):** Wav2Vec2 IEMOCAP Emotion Classifier

References

- [1] An, K., Chen, Q., Deng, C., Du, Z., Gao, C., Gao, Z., Gu, Y., He, T., Hu, H., Hu, K., Ji, S., Li, Y., Li, Z., Lu, H., Luo, H., Lv, X., Ma, B., Ma, Z., Ni, C., ... Wang, Y. (2024). *FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs*. arXiv. <https://doi.org/10.48550/arxiv.2407.04051> (Cited by: 108)
- [2] Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). *wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations*. arXiv. <https://doi.org/10.48550/arxiv.2006.11477> (Cited by: 10714)
- [3] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). *Robust Speech Recognition via Large-Scale Weak Supervision*. arXiv. <https://doi.org/10.48550/arxiv.2212.04356> (Cited by: 9973)